

MinVol three-dimensional contact bound

Repository source: README.md

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Release: [v1.1.0](#) (September 26, 2026) **Author:** Bruce J. Swihart **Status:** AI-assisted, non-peer-reviewed computer-assisted research preprint seeking independent mathematical verification **Canonical repository:** github.com/swihart/minvol-degree3-contact-bound **AI system disclosed:** OpenAI ChatGPT (GPT-5.6 Sol Pro), accessed August-September 2026

This repository is the paper, exact certificate, and reproducibility package for the proposed universal bound

$$\text{Vol}(K) \geq \frac{26221074253\pi}{200000000000} d^3 = 0.41187967121228637754323153860 \dots d^3$$

for every convex body K in $\mathbb{R}^3$ of constant width d .

The package contains a standalone exact-rational certificate, independent binary64 Python and base-R audits, ten reconstruction tests, seven adversarial proof-object mutation tests, a ten-page manuscript, and paired Markdown/PDF documentation. Bruce J. Swihart is the sole named author; no institutional affiliation is asserted. The paper and prose are licensed CC BY 4.0, while the software and build infrastructure are licensed MIT.

Paper

- [Reference PDF](#)
- [LaTeX source](#)
- [Paper build and reconciliation guide](#)
- [Rendered paper-guide PDF](#)

The manuscript explains the geometric reduction, the two circumradius branches, and the finite certificate architecture. Every theorem-critical constant and the complete lower-branch partition are generated from the machine-readable certificate rather than copied manually.

Publication metadata

- **Author:** Bruce J. Swihart
- **Institutional affiliation:** none asserted

- **Correspondence and verification reports:** <https://github.com/swihart/minvol-degree3-contact-bound/issues>
- **Version:** v1.1.0
- **Release date:** September 26, 2026
- **Versioned release:** <https://github.com/swihart/minvol-degree3-contact-bound/releases/tag/v1.1.0>
- **License:** CC BY 4.0 for paper/prose; MIT for software/build infrastructure
- **Citation metadata:** [CITATION.cff](#)
- **Licensing text:** [LICENSE](#)

Verification record

Before the final version transition, the release-candidate commit `892d4bc6fad07950e78da66e45b95063a6415af6` passed a three-job hosted workflow covering exact/Python verification, independent base R, and paper/document builds with Poppler preflight. A separate fresh clone of that commit reproduced the complete Python and R replay under Python 3.14.7, NumPy 2.3.5, and R 4.6.1; rebuilt the ten-page paper and all 22 Markdown/PDF pairs; and ended with an empty `git status --short`.

The archived evidence and exact transfer hashes are in [CLEAN_CLONE_CHECK.md](#) and [certificate/evidence/rc1_remote_audit.json](#). The final v1.1.0 tag is required to pass the same three hosted jobs before the GitHub Release is published. These are reproducibility checks, not peer review or external mathematical verification.

Reviewer path

1. Read the [paper PDF](#).
2. Read the [certificate status](#) and [certificate guide](#).
3. Verify the checksum manifests.
4. Run the exact and independent Python paths with `./run_python_checks.sh`.
5. With base R installed, run the complete replay with `./run_all.sh`.
6. Consult the proof ledger, source ledger, and trust-boundary document.

Replay

Install the pinned Python dependency:

```
python3 -m pip install -r requirements.txt
```

Run the exact verifier, binary64 audit, ten reconstruction tests, seven adversarial mutation tests, exact-constants and release-document checks, release metadata preflight, and paper/certificate reconciliation:

```
./run_python_checks.sh
```

Expected markers include:

```
MINVOL UNIVERSAL CONTACT-BOUND CERTIFICATE: EXACT
MINVOL UNIVERSAL CONTACT-BOUND BINARY64 AUDIT: PASS
MINVOL ADVERSARIAL MUTATION SUITE: PASS
MINVOL EXACT-CONSTANTS APPENDIX: CURRENT
MINVOL REMOTE-EVIDENCE CHECK: PASS
MINVOL RELEASE-DOCUMENT CHECK: PASS
MINVOL CITATION METADATA CHECK: PASS
MINVOL METADATA PREFLIGHT: PASS
MINVOL RELEASE PREFLIGHT: PASS
MINVOL PAPER-CERTIFICATE RECONCILIATION: PASS
```

With base R installed, run all language paths:

```
./run_all.sh
```

After changing a protected source, transcript, or reference artifact, refresh and verify the checksum manifests:

```
python3 refresh_checksums.py
```

```
python3 refresh_checksums.py --check
```

Build documentation and release assets

Build the paper without overwriting its committed reference PDF:

```
./build_paper.sh
```

Build local PDFs for every Markdown source under `build/markdown/`:

```
./build_markdown_pdfs.sh
```

The committed Markdown/PDF pairs are under `rendered/markdown/`. Their pairing and PDF preflight checks run in continuous integration.

Preview and verify the deterministic release-asset set before tagging:

```
./build_release_assets.sh --preview
```

```
./verify_release_assets.sh
```

From the exact tagged checkout, omit `--preview`. The resulting release bundle contains the paper PDF, paper source, standalone certificate, paired documentation, replay records, release notes in Markdown/PDF, a machine-readable release record, and one SHA-256 manifest.

Proof architecture

The proof splits the normalized circumradius interval at

$$R_* = \frac{305695601}{500000000} = 0.611391202.$$

Below the split, a gap-free exact radial-propagation certificate has unique weakest terminal row 03B02B. Above the split, four balanced circumsphere contacts force a 98,306-ray polyhedral core. An exact determinant transfer, a dominant-edge coordinate-width estimate, and three optimized antipodal cap pairs produce a near-Jung lower bound that is the universal bottleneck, while both split terminal rows clear the theorem coefficient. Exact integer and rational arithmetic supplies the finite proof authority; binary64 Python and base R are independent numerical audits.

Documentation

The editable sources are Markdown. Checksummed reference PDFs mirror the same repository-relative tree under `rendered/markdown/`.

Status, release, and provenance

- Verification status: [Markdown](#) | [PDF](#)
- Reproducibility guide: [Markdown](#) | [PDF](#)
- Clean-source record: [Markdown](#) | [PDF](#)
- AI assistance and provenance: [Markdown](#) | [PDF](#)
- Release notes: [Markdown](#) | [PDF](#)
- Release checklist: [Markdown](#) | [PDF](#)

Certificate documentation

- Certificate specification: [Markdown](#) | [PDF](#)
- Trust boundary: [Markdown](#) | [PDF](#)
- Claims and limitations: [Markdown](#) | [PDF](#)
- Independent-audit record: [Markdown](#) | [PDF](#)

- Certificate guide: [Markdown](#) | [PDF](#)
- Certificate status: [Markdown](#) | [PDF](#)

Mathematical proof documentation

- Mathematical overview: [Markdown](#) | [PDF](#)
- Claim dependencies: [Markdown](#) | [PDF](#)
- Proof ledger: [Markdown](#) | [PDF](#)
- Internal proof audit: [Markdown](#) | [PDF](#)
- Source ledger: [Markdown](#) | [PDF](#)
- Appendix A, exact constants: [Markdown](#) | [PDF](#)
- Appendix B, certificate architecture: [Markdown](#) | [PDF](#)
- Appendix C, replay transcripts: [Markdown](#) | [PDF](#)
- Paper guide: [Markdown](#) | [PDF](#)
- This README: [Markdown](#) | [PDF](#)

Citation and license

Use [CITATION.cff](#) for the preferred preprint citation. The paper and prose documentation are licensed under CC BY 4.0; software and build infrastructure are licensed under MIT. See [LICENSE](#) for the full split-license notice.

Scope

The package proposes a universal lower bound. It does not identify a minimizing body, prove sharpness or equality, or prove the Meissner conjecture. The Meissner value is an explicit-body comparison value, not a universal minimum proved here. The result has not been externally reproduced or peer reviewed. The finite certificate is not a Lean theorem-prover formalization; here, **exact** means exact integer and rational arithmetic checked by the included programs.

The earlier public MinVol spectral announcement remains a separate repository: [swihart/minvol-degree3-spectral-bound](#). Further universal refinements are under investigation but are outside this version.